

Arturo Cardenas-Blanco
Expanded by Renat Yakupov
DZNE-Magdeburg
renat.yakupov@dzne.de

DELCODE – Cortical and subcortical thicknesses and volumes
FULL BASELINE RELEASE – DELCODE1000

Thickness and volume measurements obtained after processing the T1-weighted images using Freesurfer version 6.0. This document was written to briefly explain the output variables contained in the files listed below.

Filenames:

lh.aparc.thickness_2019-10-01.csv
rh.aparc.thickness_2019-10-01.csv
lh.aparc.volumes_2019-10-01.csv
rh.aparc.volumes_2019-10-01.csv
aseg_stats_2019-10-01.csv
HC_subfields_2019-10-01.csv
Freesurfer_labels.txt
Freesurfer_proc_info.txt

Overview:

	Cortical ROIs		Subcortical ROIs		Hippocampus
Measure	Thickness (mm)	Volume (mm ³)	-	Volume (mm ³)	Volume (mm ³)
Filename	?h.thickness...	?h_volumes..	-	aseg_stats...	HC_subfields...

Cortical ROIs

The “l” and “r” prefixes in the filenames (lh.thickness... and rh.thickness...) are used to indicate left and right hemisphere respectively.

In these files, the first row is a list of parcellation names. The first column is the subject name. If the measure is thickness then the last column is the mean cortical thickness.

More information about the cortical parcellation can be found here:
Automatically Parcellating the Human Cerebral Cortex, Fischl et al., (2004).
Cerebral Cortex, 14:11-22

Subcortical ROIs

The “aseg_stats...” file contains a table in which each line is a subject and each column is a segmentation. The values are the volume of the segmentation in mm³. The first row is a list of the segmentation names. The first column is the subject name. The estimated intracranial volume (eTIV) is printed as the 2nd to last column, and BrainSegVol is the last column.

More information about the subcortical parcellation can be found here:

<https://surfer.nmr.mgh.harvard.edu/ftp/articles/fischl02-labeling.pdf>, Fischl et al., (2002). Neuron, 33:341-355.

Hippocampus

The volumes reported in this text file “HC_subfields...” have been obtained using an advanced segmentation tool also part of Freesurfer 6.0. This tool generates an automated segmentation of the hippocampal subfields based on a statistical atlas built primarily upon ultra-high resolution (~0.1 mm isotropic) ex vivo MRI data. This new method can take advantage high-resolution, dedicated images when available (typically, but not necessarily, T2 weighted), and solves a number of limitations of the in vivo atlas that was distributed with FreeSurfer 5.1-5.3.

To obtain a more reliable segmentation of the Hippocampus, we have used two MR images, the T1-weighted and the high resolution T2-weighted image. If you are interested in HC volume measurements, please use these values rather than the ones stored in “aseg_stats...” file. This HC segmentation is more reliable and therefore produces more accurate results.

More information about the Freesurfer 6.0 hippocampus segmentation can be found here:

<http://www.nmr.mgh.harvard.edu/~iglesias/pdf/subfieldsNeuroimage2015preprint.pdf>, Iglesias, J.E., Augustinack, J.C., Nguyen, K., Player, C.M., Player, A., Wright, M., Roy, N., Frosch, M.P., McKee, A.C., Wald, L.L., Fischl, B., and Van Leemput, K. Neuroimage, 115, July 2015, 117-137.

FreeSurfer labels

Some values have cryptic labels. For those please refer to Freesurfer_labels.txt file, which contains expanded names for those abbreviations.

FreeSurfer processing information

Unfortunately, the input data was not identical for all subjects. There were several differences between input files. These differences are documented in FreeSurfer_proc_info.txt file. For each subject, there is additional information as follows:

T1T2 column – for various reasons, the high-resolution T2-weighted image was not acquired for several subjects. In those cases, hippocampal subfields segmentation was carried out using just the T1-weighted MPRAGE image, which is suboptimal. For most subjects, this field has T1T2 value, indicating that high-resolution T2-weighted image was used. For some, this field equals T1, which

means no T2-weighted image was used.

PAT column – most of the T1-weighted MPRAGE images were acquired using acceleration factor of 2, some, however, were acquired without acceleration. This column contains the acceleration factor. Acquisition acceleration reduces image signal-to-noise ratio by the square root of acceleration factor. However, it can potentially have higher image quality due to less subject motion during the reduced acquisition time.

T1 column – not all images were acquired with the same post-acquisition processing options. This column contains information about post-acquisition processing steps carried out on scanner. NORM stands for intensity normalization (bias-field correction), DIS2D stands for distortion correction in phase-encoding direction, DIS3D stands for distortion correction in the 2nd phase-encoding direction in case of 3D sequences, and ND stand for non-distortion-corrected. Distortion correction improves coregistration steps carried during processing, therefore the segmentation and parcellation results are more reliable. However, this kind of acquisition doesn't suffer from strong distortions (especially at 3T magnetic field).

T2 column – same as for T1 column, except for the high-resolution T2-weighted image used for hippocampal subfields segmentation.